Group 4 Reflection Journal

ITAI 2376 | Diffusion Models Midterm

Esther Odaibo

My contribution was preparing the assessment report covering diffusion concepts, model architecture, training analysis, CLIP evaluation, and practical applications. Analyzing our notebook results helped me understand that decreasing training loss does not always produce recognizable or correctly labeled images. I also learned that high CLIP scores require careful interpretation and should be supported by visual inspection and additional evaluation. I identified limitations and proposed improvements for the team to consider. In future projects, I would participate earlier in implementation and help compare changes through controlled experiments.

King Owusu

Data selection, MNIST and loading, the U-Net, the diffusion process, training, generation, and CLIP evaluation. I wrote the assessment answers, the analysis report, and my learning notes on the model components, working from the notebook's outputs. What went well? Splitting the work this way let us each focus on one thing. Reading James's outputs closely showed me things the loss number alone did not, such as the schedule ending before pure noise. What was difficult? The generated digits were not readable, which meant my write-up had to be honest about a weaker result than I expected. It was also hard to separate "the model trained well" from "the model generates well," since those are different things. What I learned. A low loss does not guarantee good samples. A CLIP "clarity" score is not an accuracy score. Skip connections and conditioning placement matter a lot to what a U-Net can do. What I would do differently. Test the model's output with a digit classifier earlier, and add the conditioning in every block from the start.

James Delgado

My main contribution to this project was working through the diffusion notebook from beginning to end, including dataset preparation, model components, training, image generation, and CLIP evaluation. I also helped troubleshoot errors and added comments to explain what the code was doing. Completing the full notebook helped me see how the different parts depended on one another. The most challenging part was connecting unfamiliar Python code and mathematical ideas to the purpose of each step. I used AI extensively for explanations and debugging, and I spent about eight hours working through the notebook cell by cell. I found it helpful to ask for explanations from first principles, including where an idea came from, why it was needed, and how information moved through the model. I still need more practice writing this code independently, but I made progress in understanding the process. One important lesson was that lower training loss does not guarantee recognizable images. Our model produced visible strokes, but many samples did not resemble the requested digit. I also learned to interpret CLIP results carefully. A high weight for the “clear” description did not mean that the model had generated the correct number. Comparing the images with the reported scores helped me understand the limits of those measurements. For a future project, I would check generated samples earlier, keep saved outputs consistent with the final notebook run, and establish clearer group responsibilities at the beginning. I would also break the work into smaller experiments so I could

connect each change to its effect. My goal is to keep building enough understanding to explain, modify, and eventually implement more of the model independently.

Zain Ahmed

My part of the midterm focused on checking and organizing our group's final work. I brought the notebook, assessment report, and journals together, reviewed the PDF files, and handled the GitHub submission materials. I learned that a lower training loss does not always mean the model is producing better images. Our model learned to predict noise more accurately, but some generated digits were still hard to recognize. I also learned why CLIP's clear-image scores should not be treated as digit-recognition accuracy. The hardest part for me was making sure the different files matched the assignment requirements and that the notebook results were represented accurately. This project showed me that reviewing results and preparing a clear submission matter too. Next time, I want to join the coding work earlier so I can learn more about how the model is trained and improved.